

Effects of Video Self-Modeling on Spontaneous Requesting in Children with Autism



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Abstract: Video self-modeling (VSM) is a promising intervention to teach new skills and improve the use of existing skills in young children with autism. VSM includes observation and imitation of one's self on videotape that records specific desirable child behaviors. The purpose of this study was to test the effectiveness of VSM for training young children with autism to make spontaneous requests in school settings. Four young children with autism participated. Experimental control was demonstrated using a multiple-baseline design across participants. Introduction of VSM led to a large increase in requesting behavior in all four children. VSM was effective in causing an increase in spontaneous requesting in young children with autism.

Children with autism typically display great difficulties in social-communication skills. Poor social communication includes poor social interactions, such as poor imitation skills, and lack of eye contact, and lack of responsiveness to people. Limited or nonexistent communication skills, such as lack of turn-taking skills, problems with speech/communication, or poor or limited use of appropriate gestures, are also common. Finally, delayed, restricted, repetitive, or unusual responses to stimuli, such as squealing, screaming, and hand flapping are also common examples of poor social-communication skills.

This lack of social-communication skills is also a significant problem for children with autism. Social-communication skills rely on reciprocal interactions between individuals. Lack of skill in this area often deters other children from beginning or maintaining communication and friendships with children with autism. Social-communication skills are an important aspect of human interaction. Deficits in eye contact, turn taking, gestures, or verbalizations that indicate requesting objects or action/attention, responding to others, or protesting (e.g., refusing to participate in an interaction) are perceived by peers, family members, and other individuals as significant limitations in the social-communicative behavior. Difficulties in social-communicative behaviors can impede children's reciprocal social-communicative interactions (Wetherby, 1998) and subsequent ability to function in society (McGee & Daly, 1999). A lack of skills in this area in children with

autism thus is a major concern. An important goal in interventions for children with autism is to increase social-communicative development in order to decrease the problems often produced by social avoidance or awkwardness.

One effective strategy to aide young children in learning social-communication skills is to model the behavior of individuals around them. This process has been termed *observational learning* or *modeling* (Bandura, 1977). *Video self-modeling* (VSM) is a form of modeling that allows individuals to observe their own behavior. In other words, the person acts as his or her own model (Hosford, 1981). VSM research, which was based on the Bandura modeling research, began approximately 30 years ago, and about 150 studies of VSM have been published (Dowrick, 1999). Dowrick, and Meharg and Woltersdorf (1990), reviewed the VSM literature and found many demonstrations of the use of VSM that improved personal and social adjustment, physical skills, and vocational training. In both articles, substantial gains were reported across all target behaviors both immediately following intervention and after intervention was withdrawn. VSM thus is a promising and acceptable methodology for improving prosocial behaviors.

Zihni and Zihni (1995), parents of a nonverbal 5-year-old with autism, used VSM to teach vocabulary to their child. They reported that their child increased her vocabulary by 250 words in 6 months, thus making VSM effective in this area. Zihni and Zihni suggested several reasons

for VSM's effectiveness: Many children with autism are uncomfortable with human interaction, many of these children are highly visual, video is predictable and controllable, and video allows for the filtering out of extraneous variables. In spite of the possible efficacy of VSM, only three studies that used VSM in children with autism have been identified (Buggey, Toombs, Gardener, & Cervetti, 1999; Charlop & Milstein, 1989; Krantz, MacDuff, & McClannahan, 1993). The purpose of this study therefore was to test the effectiveness of VSM as a strategy for improving social-communication skills by increasing the frequency of spontaneous request behaviors for young children with autism.

Method

PARTICIPANTS

Information about the research project was disseminated to recruit volunteer families for the study. Families were interviewed and consent forms were explained and signed. The participants were four preschool children ages 3 years to 6 years with a diagnosis of autism (American Psychiatric Association, 1994). The children had a variety of language capabilities, as noted on their Individualized Education Programs (IEPs). Spontaneous requests by these participants were either minimal or nonexistent.

Participant 1 was a 5½-year-old boy who had been diagnosed with autism at the age of 3 years. His language skills were on target for his age, although he still had difficulty making spontaneous requests, answering "wh" questions, and using the pragmatics of social language. Participant 2 was a 4½-year-old boy who had been diagnosed with autism at the age of 2 years. His language skills were approximately 1 year behind his chronological age. He also had difficulty making spontaneous requests, answering questions, and using the pragmatics of social language. Participant 3 was a 4-year-old boy who had been diagnosed with autism at the age of 3 years. His language skills were at the preverbal level, with only vocalizations and gestures used for communication. He had difficulty making spontaneous requests verbally but was beginning to use a communication board. Participant 4 was a 5-year-old boy who had been diagnosed with autism at the age of 3 years. His language skills were at the 2-year-old to 3-year-old level. He had difficulty making spontaneous requests, had a limited vocabulary, and was just beginning to use two-word phrases.

TARGET BEHAVIOR

The target behavior was named *spontaneous requesting* (SR), which was operationalized as independently asking for an

object or action without assistance. The frequency of SR was the dependent variable.

MATERIAL

For each child, a videotape was made of his own SR behavior. Making each videotape involved eliciting and videotaping SR behaviors from the child and editing the videotape to show only desired request behaviors. Orchestrated play sessions conducted at each child's own home setting were videotaped to provide footage that could be edited to create training tapes for intervention. The sessions were tape-recorded using a digital video camcorder balanced on a stationary tripod at least 6 ft. from the play situation. Videos were edited on the computer using a digital editing program that was set up with enough memory to edit videotape and transfer it to a VHS tape for participant use. Videotapes were edited to delete adult prompting of requesting behavior as well as negative child behaviors. The completed tapes were approximately 5 minutes in length. Edited tapes showed only requests that appeared as unprompted and spontaneous. Occasional requests by the adult were included, when appropriate, to maintain turn taking and conversation with the participant. Edited tapes were completed by adding an opening page and then copying the digital tape to VHS tapes for viewing on a family videocassette player.

PROCEDURE

Baseline data were collected in the school settings, where 30-minute play sessions were observed. Probes were administered to determine the baseline for each participant. Adult prompters, who were behavior therapists already working with the families, then were asked by the families to participate in individual training sessions with the investigator. Three 30-minute training sessions for adult prompters were conducted. Training sessions used an explicit instruction model, with adults taking turns acting as the participant for the first two 30-minute sessions and as the prompter for the last 30-minute session until they could elicit 10 request opportunities in each of the three 30-minute sessions. Prompters were trained to count opportunities and were considered fully trained when they could spontaneously replicate the orchestrated play model without direction with 100% accuracy for all 10 request opportunities in each of the three sessions.

Prior to this study, the child participants were trained in requesting using discrete trial drills modeled after the *Teach Me Language* manual (Freeman & Dake, 1996). The participants were all able to comply with requests, when motivated, prior to beginning the study. Although they used spontaneous requests on a minimal basis, all the par-

ticipants needed constant prompting to make verbal and nonverbal requests in the school setting.

Material to create VSM training tapes was collected during one 30-minute orchestrated play session in the home setting. Participants were asked if they would help make a movie in which they would star and with what toy or activity they would like to play for the movie. Participants were allowed to choose toys or activities that were of interest to them, such as a range of board games (Candyland™, Hi Ho Cherrio™, Hungry Hippo™), play dough, action figures, trains (Thomas the Tank Engine™ sets, puzzle play), and quiet time reading of favorite storybooks. Parents had also been asked to list toy and play choices for their children prior to the study. Videotaping began as adult prompters and children began orchestrated play sessions in the home settings.

The format for orchestrated play to create training tapes used the following steps:

1. The adult prompter asked the child to play by saying, "Let's play."
2. The adult prompter asked the child to choose a toy or activity for play by saying, "What would you like to play?"
3. The adult prompter began to set up the play activity.
4. The adult prompter withheld pieces of the play activity (materials) so that the participant needed to request the missing materials.
5. The adult prompter modeled request behaviors by asking, "May I have some play dough?" or "May I have a game piece?"
6. The adult prompter prompted the participant to ask for objects or actions by saying, "Now you say, 'Give me play dough' or 'B's turn/my turn.'"

Prompts were used continuously so that multiple examples of participant requesting would appear on the training videotape. VSM training tapes with the prompts and negative behaviors edited out were made for participant viewing.

Participants were then asked to watch their own 5-minute videotape at home one time each day for 5 consecutive school days within 60 minutes of school attendance. Parents were asked to observe/videotape their children watching their videos to document that the children were attending to the videos as well as to note any possible negative reactions to the videos. The parents were instructed to stop the tape if they or the participant was uncomfortable with it. No adverse reactions were noted. Some jiggling movements on the part of the children were noted and judged by the parents to be indicative of discomfort, but these parents did not stop the video viewing. This occurred only during the first viewing of the tapes.

Data were collected in the school settings during the intervention and maintenance phases of the study. All adults in the school were asked not to provide any prompting during data collection in these phases.

DATA COLLECTION AND EXPERIMENTAL DESIGN

Observations were tallied using event recording for spontaneous requests only. Classroom assistants, teachers, and parents were asked not to prompt during the intervention and maintenance phases of the study.

The investigator trained observers on the operational definition of SR and also on how to use event recording during observations of SR behaviors. Observers were considered fully trained when 100% agreement was reached by all observers on three 15-minute video samples of a child with autism. Interobserver agreement on SR behavior using event recording was assessed in 50% of the baseline phase, 60% of the VSM intervention phase, and 100% of the maintenance phase. The agreement formula used was the number of agreements (occurrence only) divided by the total number of agreements (occurrence only) plus the number of disagreements (occurrence only), multiplied by 100%. The interobserver agreement for all phases of the study was 100%. A single-subject design using multiple baselines across subjects was employed to demonstrate experimental control of the participant's behavior.

Results

All the data are presented in Figure 1. At baseline, Participant 1 engaged in a mean of 0.83 SRs (range 0–3). During intervention, his SR behaviors demonstrated a variable but increasing trend, with a mean of 10.2 SRs (range = 5–19). Maintenance data were collected for three sessions over a 6-week period. During maintenance, Participant 1 maintained a high frequency of SRs, with a mean of 23.6 (range = 23–25).

At baseline, Participant 2 engaged in a mean of 2.7 SRs (range = 0–5). During intervention, he demonstrated a variable but increasing trend in SRs, with a mean of 17.5 (range = 11–26). During maintenance over a 4-week period two data points were collected. The mean number of SRs was 20.5 (range = 19–22).

At baseline, Participant 3 engaged in a mean of 1.0 SRs (range = 0–4). During intervention, he demonstrated a variable but increasing trend in SRs, with a mean of 12.6 (range = 7–23). One data point was collected during a 2-week maintenance phase. The mean frequency of spontaneous requests was 21 for that occasion.

At baseline, Participant 4 engaged in a mean of 1.2 SRs (range = 0–5). During intervention, the mean number of SRs was 13.2 (range = 2–27). Participant 4 was unable to continue with the study through the maintenance phase.

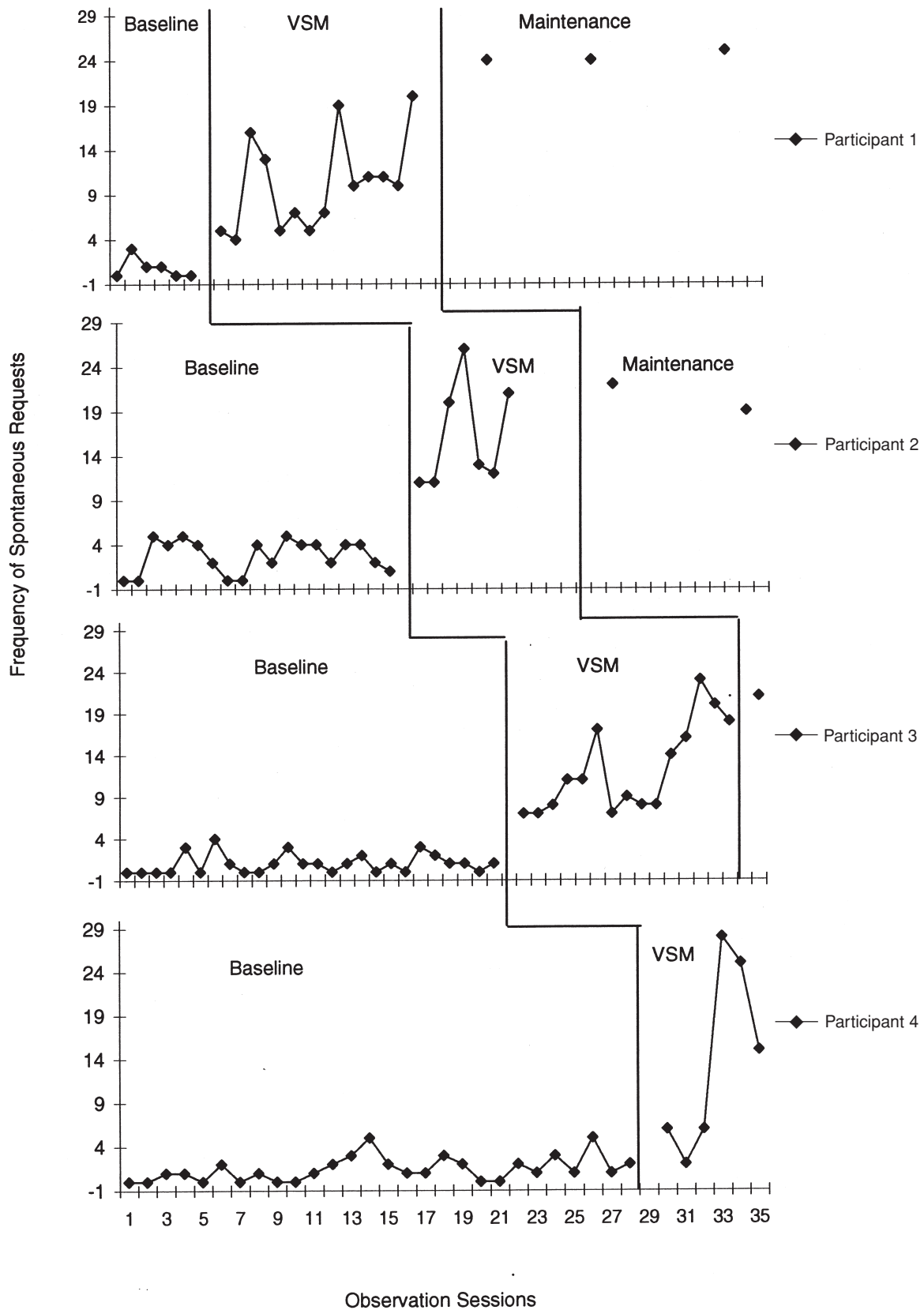


Figure 1. Video self-modeling multiple-baseline data.

Discussion

This study demonstrated that VSM caused a marked increase in SRs in preschool settings for each participant. Only Participant 4 demonstrated a delay in increased frequency of SRs after the introduction of VSM. It should be noted that he was not interested in video viewing or in seeing himself in a movie; however, he still had increases in SRs during intervention. Furthermore, the maintenance data for the first three participants indicated that these changes were maintained over a 2- to 6-week period. These results thus demonstrate that VSM was effective in increasing the frequency of these participants' SRs. This study also demonstrated generalization of the skill from home to school settings. Generalization was not intentionally programmed into the VSM tapes, but the skill was evidenced in school settings, perhaps because the activities chosen for the VSM tapes were similar to activities in the school setting. The adult prompters who appeared with the participants in the VSM tapes did not see the participant children in school; thus, neither the VSM video viewing settings nor the adult prompters were likely factors in generalization.

Two issues must be regarded when considering the results of the study. First, the participants had been previously trained in the target behavior, but they had demonstrated SR on a minimal basis in home and school settings. The target behavior thus was not a new skill but was one that was demonstrated at a low frequency. Second, three of the four participants liked to watch videotapes and agreed to participate in making a videotape, suggesting that the VSM intervention was highly motivating for these three participants.

Three previous studies using VSM in children with autism reported improvements in social-communication skills (Buggey et al., 1999; Charlop & Milstein, 1989; Krantz et al., 1993). Although limited in number, these studies on the effectiveness of VSM have yielded positive results. Replication of this study using preschoolers of varying ages and in different settings is necessary to determine if VSM is equally effective at different ages or stages of development. None of the studies, including this investigation, separated the effects of VSM to generalize across different types of individuals in different settings. Future research might include an analysis of SRs to peers versus adults, as well as changes in the behavior of peers and adults to the child with autism, following increases in SRs.

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